

Co-Formation Games — Unifying Network Formation and Opinion Dynamics

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Abstract—This report proposes a novel model combining network creation studies and opinion dynamics. After introducing each Network Formation Games (NFGs) and Social Influence Models (SIMs), I compose a model allowing for co-evolution of opinions and networks and containing both NFGs and SIMs as special cases. Other sections consider basic dynamics, optimal states, and research directions of these Co-Formation Games (CFGs), as well as showcase an Julia module implementing CFGgames. Overall, the model offers a promising basis for future research projects, while broadening the perspective of NFGs and SIMs.

Index Terms—Collective dynamics, Game theory, Network formation games, Network theory (graphs)

I. INTRODUCTION

Network dynamics plays an ever-important role in describing the modern world, with connective plasticity and local optimizations enabling the emergence of global order. Since this framework can describe a vast variety of phenomena, researchers employed numerous approaches — often multi-disciplinary ones — to analyze these systems. Specifically, networks of human-like agents have been subject of studies ranging from mostly theoretical analysis to phenomenological social science works [1], [2].

Even as the sheer numerousness has inspired some authors to question each model's marginal benefit, considering strangs of research showcases a lack of holistic perspectives [1]: As a subfield of game theory, Network Formation Games (NFGs) study how self-interested agents create and drive the structure of networks [3]. In contrast, Social Influence Models (SIMs) focus on inner states of nodes (opinions) diffusing through it [2]. In the case of social networks, this raises a tension: As opinion propagation and network formation can both be dynamic and influence each other, results from studies regarding only NFG or SIM do not necessarily generalize to the case of simultaneous dynamics.

This is an important caveat, as real social network behaviour rarely reduces to the semi-static treatment implied by NFGs and SIMs. Yet, even though there are works trying to combine some aspects of the other's influence, they often do so as a small part of the original model's main mechanism (e.g., [4], [5], [6]). Moreover, there is still no work unifying both approaches in a way sheltering results from both disciplines.

This report explores a model which allows topology and opinions to influence each other, while including NFGs and SIMs as special cases. After presenting NFGs and SIMs, this work presents and justifies each building block of the model, before assessing some basic traits with regard to possible dynamics, optimality and a simulation module implemented in the Julia programming language.

II. BACKGROUND

A. Network Formation Games

Introduced as a subfield of Game Theory, Network Formation Games (NFGs) consider agents acting strategically to maximize personal utilities / minimize costs [3]. In NFGs, the action agents can decide on is whether or not to sponsor connections to an other agent. An action profile a — storing every single agent i 's decisions a_i — consequently produces a graph $G(a)$ in which nodes represent agents and edges [7].

(Since agents are always treated as part of a graph, the following text uses *agent*, *node* and *vertex* interchangeably, unless noted otherwise.)

The cost function $c(a_i)$ which each agent wishes to minimize imposes costs α_c on each connection sponsored, but encourages connectedness by punishing a distance measure of i in the graph, $d_i(G)$. When agent i sponsors an edge to j , we set $a_{ij} = 1$, and $a_{ij} = 0$ otherwise, so that

$$c_{i(a_i)} = d_i(G) + \alpha_c \|a_i\| \quad (1)$$

returns the cost an agent incurs, with $\|\cdot\|$ being the L1 norm.

Because agents are unaware of the other game participants' actions (knowledge of a and the resulting network $G(a)$ is only implicit in (1)), no explicit coordination is possible. Instead, step-wise adaptations of a_i are taken to reduce costs, until there is no single sponsorship change left which improves costs. These final states a^* are so-called Nash equilibria (NEs). They correspond to the concept of local stability, and a big portion NFG-related research devotes itself to analyzing the set of NEs and NE traits.

Since costs are minimised locally and step-wise, a configuration which minimizes the total cost

$$c(a) = \sum_{i=1}^n c_i(a_i) = d(G) + \alpha_c \|a\| \quad (2)$$

may be unreachable:

$$a^{\text{opt}} = \arg \min_a c(a) \leq \arg \min_{a^*} c(a^*). \quad (3)$$

This relates to the Price of Anarchy, which presents how costly selfish strategies end up being relative to game Γ 's optimum costs profile,

$$\text{PoA}(\Gamma) = \max_{a^*} \frac{C(a^*)}{C(a^{\text{opt}})}. \quad (4)$$

Overall, while there are many studies regarding optimality considerations for NFGs in terms of topology [7], [8] and some research weighing costs by categorical opinions [4], NFGs are yet to include explicit information propagation their dynamics.

- Varieties:
- Laterality – Directedness

B. Social Influence Models

Social Influence Models (SIMs) investigate how agents interacting with each other may adapt their behaviour due to those interaction, given some theoretical simplifications.

(- Sidenote: Often used interchangeably with opinion dynamics, sorry Michael)

- Basic idea: Agents have opinions. Interaction = opinion exchange => Consequence
- Historically, most models are random interactions, i.e., have no topology.
- Some simple models
 - deGroot: Averaging real numbers – Ising model: we are all discrete spins – Modern stuff: Homophily, Similarity. — Structure implied, but not specific. – Bounded confidence et al. Because convergence was boring.
- Criticisms: No model captures true human behaviour so far. Duh.
- For our purpose: Edging closer to holistic model. No wish to capture human spirit or anything.
- Rather, see “opinion” as label for local information connected to edge; “social” as nodes influence each other due to “personal” information.

III. THE MODEL — CO-FORMATION GAMES

- This section shows the choices. Considered alternatives are in the appendix.
- Unilateral, Undirected
- TBW
- (basically write down notebook page)

A. Definitions & Equations

TABLE I
LIST OF VARIABLES USED IN THIS DOCUMENT

Variable	Symbol	Domain
Number of nodes	n	$\mathbb{N}$
Node of concern	i	$\{1, \dots, n\}$
Set of i 's neighbours	K_i	$\mathbb{N}^{ K_i }$
Network graph ¹	G	$\{0, 1\}^{n \times n}$
Opinion	x_i	$\mathbb{R}$
Opinion profile	x	$\mathbb{R}^{n \times n}$
Action	a_i	$\{0, 1\}^n$
Action profile	a	$\{0, 1\}^{n \times n}$
Costs per edge	α_c	$[0, \infty[$
Pairwise distance	d_{ij}	$\mathbb{N}$
Network valuation	v	$] - \infty, 0]$
Network expenses	f^c	$] - \infty, 0[$
Cost function	c	$] - \infty, 0[$
Communication rate	r_c	$[0, \infty[$
Formation rate	r_f	$\{0, 1\}$
Convergence threshold	ε	$[0, \alpha_c]$

Up for discussion

where a norm $\|\cdot\|$ is the L0 norm, returning the count of non-zero elements.

a) *Equations:*

- Cost function

$$c_i = v_i(a) + f^c(a) \quad (5)$$

- Valuation function

$$v_i(a) = - \sum_{j=1, \dots, n} d_{ij} \quad (6)$$

- Network expenses:

$$f_i^c(a) = -\alpha_c \|a_i\| - \frac{1}{\|K_i\|} \sum_{j \in K_i} |x_i - x_j| \quad (7)$$

b) *Algorithm:*

¹The graph is represented as an adjacency matrix.

TABLE II
THE PLANETS OF THE SOLAR SYSTEM AND THEIR AVERAGE DISTANCE FROM
THE SUN

Planet	Distance (million km)
Mercury	57.9
Venus	108.2
Earth	149.6
Mars	227.9
Jupiter	778.6
Saturn	1,433.5
Uranus	2,872.5
Neptune	4,495.1

```

1 Initialise  $t = 1, c$ 
2 while  $(t < t_{\max} \wedge \varepsilon < \Delta c)$ 
3   if  $\text{mod}(t, r_c) = 0$  then
4     select random node  $i$ 
5      $x_i$  arrow.l get opinion update  $x_i^u$ 
6   end if
7   if  $r_f = 1$  then
8     select random node  $i$ 
9      $a_i$  arrow.l get action update  $a_i^u$ 
10  end if
11   $t += 1$ 
12   $c(t)$  arrow.l get new costs  $c_t$ 
13   $\Delta c = c_{t-1} - c(t)$ 
14 end while

```

Algorithm 1. Dynamics of our synchronous move game

IV. INTRODUCTORY EXAMPLES

A. Cost function behaviour

- Copy some of your paper drawings

B. New optimal costs

- Examples of chains vs star

C. Some simulations

V. OLD STUFF FROM THE TEMPLATE



Fig. 1. A circle representing the Sun.

In Fig. 1 you can see a common representation of the Sun, which is a star that is located at the center of the solar system.

In Table II, you see the planets of the solar system and their average distance from the Sun. The distances were calculated with (5) that we presented in

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